SA02 ASSESS AND MAINTAIN AIR TREATMENT SYSTEMS | O (MAX: 1 PT)

Intent: Mitigate risks from indoor contamination and pollution sources such as infectious disease particles and volatile organic compounds (VOCs).

Summary: This feature requires the projects to inventory air filters and other treatment devices to ensure proper maintenance.

Issue: Building materials, furnishings (e.g., carpets and furniture finishes), fabrics, cleaning products, personal care products, adhesives, solvents and air fresheners can all emit VOCs or semi-volatile organic compounds (SVOCs) into the indoor environment.[216,217] VOCs include benzene, formaldehyde and other chemical compounds, which at high concentrations can lead to irritation of the nose and pharynx and have been associated with leukemia and Nasopharyngeal cancer.[218,219] Health effects can also include damage to the liver, kidneys and central nervous system.[220] Additionally, particles exhaled by infected individuals that contain air airborne diseases such as COVID-19 can remain suspended several hours or longer and be recirculated to through the ducts of the building.[214,221,222]

Solutions: Air can be treated to remove contaminants. Carbon filters remove VOCs and ozone from the passing air.[223,224] HEPA or near-HEPA filters can help remove virus particles, since the virus often travels as part of larger particles.[213,225] UVGI systems can also be effective, both when irradiating the upper portion of the room or when placed in the air ducts, so long as they are powerful and/or the air speed is slow enough to provide sufficient UV dose.[213,226] For optimal performance, air filtration systems need to be maintained according to the manufacturer's instructions.

Part 1 Part 1 (Max: 1 Pt)

For All Spaces:

Project provides an inventory of all filters and UVGI equipment currently used to treat the air in the following locations (if any):

a. Ducts and air handling units.
b. Fan coil units.
c. Standalone air cleaning devices.

The following requirements are met:

a. A qualified engineer provides the project with an assessment of at least one of the following for occupiable areas.
    1. The highest efficiency of media or other particle filters (particularly for recirculated air, if any) that can be installed with the current mechanical system.
    2. The capacity of the current mechanical system to utilize UVGI equipment.
    3. The quantity of standalone air purifiers required to serve the space.
b. Project provides one of the following:
    1. Conditions under which project will install at least one of the types of treatment systems assessed.
    2. A timeline for the installation of at least one of the types of treatment systems assessed.

For devices identified in the System Inventory, the following requirement is met:

a. Evidence that the filters and/or UV lamps have been replaced according to the manufacturer's recommendation is submitted annually through the WELL digital platform.